

We chose Microservices Architecture for the Municipal Solid Waste Route Optimizer System.

1. Architectural Choice

We adopted a Microservices Architecture decomposing the system into decoupled, independently deployable services (Sensor Data Ingestion Service, Route Optimization Service, Driver Manifest Service, and Complaint Management Service) coordinated via a unified API Gateway and communicating through purpose-fit protocols such as gRPC, REST, and MQTT.

2. Two Scenario-Related Reasons

- **Independent Scaling for Strict Computational SLAs:** Recalculating dynamic routes for 50 collection trucks across 10,000 bins in under 45 seconds (filtering only bins over 80% full) demands intensive combinatorial optimization. Microservices enable the Route Optimization Service to scale out independently on compute-heavy nodes during peak morning dispatch without over-allocating resources to idle services.
- **Fault Isolation Across High-Throughput Streams:** Continuous high-frequency ultrasonic telemetry from 10,000 bins and sudden surges in citizen missed-pickup reports are isolated from core route planning. Network bottlenecks or memory crashes in sensor ingestion cannot cascade to degrade truck driver manifest navigation or dispatch operations.

3. Security Advantage

A centralized API Gateway enforces edge TLS termination, rate limiting, and role-based access control (RBAC) across Supervisors, Drivers, and Citizens. Network segmentation and isolated database credentials ensure that even if the public-facing Complaint Management Service were compromised, the internal fleet dispatch manifest and core routing databases remain completely protected.

4. Performance Benefit

Heterogeneous protocol optimization maximizes throughput across workloads: lightweight MQTT provides low-overhead sensor telemetry streaming, low-latency binary gRPC accelerates internal dispatch-to-optimizer calls to beat the 45-second deadline, and WebSocket streams push real-time turn-by-turn route manifests directly to drivers without continuous HTTP polling.

5. Architectural Pattern Comparison

Architecture Pattern	Scalability & Latency (45s Target)	Fault Isolation & Resilience
Layered (N-Tier)	All layers deploy together; heavy routing compute slows presentation and ingestion threads, risking the 45s SLA.	Monolithic failure domain: a memory leak or crash in sensor ingestion brings down the entire dispatch application.
Client-Server	Server forms a monolithic bottleneck for 10,000 continuous telemetry feeds and complex VRP solving.	Coupled backend: citizen complaint spikes directly compete for resources with operational truck route updates.
Microservices (Selected)	Route Optimizer scales horizontally on dedicated compute workers to guarantee sub-45s route calculation.	Independent failure boundaries: sensor surges or portal errors never interrupt active truck driver navigation.